



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VII
Course Code:		Course Name:	PM

Name of Student:	BARI ANKIT V2N09
Roll No. :	61
Assignment No.:	1
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	4
Demonstrated Knowledge	3	3
Legibility	2	2
Total	10	9

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Tejaswini PN

Signature :

Date :



Q. 1) what are the contents of project charter? who prepares and authorize the project charter?

→ A project charter is a formal document that officially authorized a project and gives the project manager authority to use organization resources.

- It is created during the initiation phase of project management.

- Contents of project charter -

(i) Project identification

- Project name

- Project id

- date of issue

(ii) Project purpose / business case

- why the project is being undertaken

- problem statement

- opportunity description

- alignment with organizational strategy.

(iii) Project objectives

- measurable goals

- expected outcomes

- success criteria

(iv) High-level scope -

- major deliverables

- boundaries of the project

- what is included and excluded.

(iv) High-level requirement -

- key functional and non-functional requirement
- major technical or business needs.

(v) Assumption and constraints -

- Resources will be available
- Technology will be compatible

(vi) High-level risks -

- major potential threats.
- uncertainties that may impact the project

(vii) Summary milestone schedule -

- key dates
- major deliverables timeline

(viii) Budget summary

- estimated project cost
- funding sources.

(ix) Key stakeholders

- sponsor
- project manager
- customer
- key team member

(x) Project manager and authority level -

- Name of project manager
- defined authority level
- responsibility boundaries

Q. 2) A startup company is developing a mobile app for online food delivery. The client initially agreed on a fixed budget and a 5 months deadline. midway through the project, the client requested additional features such as real-time tracking and AI-based recommendation system but does not want to increase the budget or extend the deadline. question - As a project manager, apply the concept of the triple constraints to this situation, explain how changes in one constraint affect others and suggest a practical solution.

→ In project management, the triple constraints refers to the three primary competing constraints -

(i) Scope (ii) time (iii) cost

$$\text{Project success} = f(\text{scope, time, cost})$$

- Scenario analysis
- initial agreement - fixed budget (cost)
 - 5-month deadline (time)
 - defined scope
- Applying triple constraints -
- scope increases - adding new features increases
 - development effort
 - testing effort
- if time remains fixed - To complete additional work
 - need add more resources
 - cannot pay overtime.
- If cost remains fixed - with budget,
 - cannot add more resources
 - cannot pay overtime
 - you cannot increase scope without affecting 2 & 3

Q 2) A mid sized engineering company has undertaken a project to design and install a rooftop solar power system for a hospital. The project involves coordination between the design team, procurement department, installation engineers, external vendors, and hospital management. Question - As a project manager, explain how you would perform your role in handling this situation. Identifying at least 10 key responsibilities you would undertake to ensure successful project completion.

→ (i) Risk management & issue resolution -

- Immediately assess impact on schedule
- communication with vendors for revised delivery date
- explore alternate suppliers or partial delivery.
- update risk register
- develop mitigation plan

(ii) Safety management

- stop work temporarily if required
- conduct safety audit
- coordination with safety officer and hospital authorities.
- implement corrective measures.
- ensure compliance with engineering and hospital safety standards.

(iii) Stakeholder communication -

- established structured weekly progress reports.
- conduct weekly review meetings.
- maintains transparency.

(iv) Cost control & financial monitoring -

- perform cost variance analysis.
- review procurement contracts.
- negotiate with suppliers.
- Identify cost-saving alternative
- Update cost forecast.